

Problem J. The Child and Sequence

Time Limit 4000 ms

Mem Limit 262144 kB

Input File stdin

Output File stdout

At the children's day, the child came to Picks's house, and messed his house up. Picks was angry at him. A lot of important things were lost, in particular the favorite sequence of Picks.

Fortunately, Picks remembers how to repair the sequence. Initially he should create an integer array $a[1], a[2], \dots, a[n]$. Then he should perform a sequence of m operations. An operation can be one of the following:

1. Print operation l, r . Picks should write down the value of $\sum_{i=l}^r a[i]$.
2. Modulo operation l, r, x . Picks should perform assignment $a[i] = a[i] \bmod x$ for each i ($l \leq i \leq r$).
3. Set operation k, x . Picks should set the value of $a[k]$ to x (in other words perform an assignment $a[k] = x$).

Can you help Picks to perform the whole sequence of operations?

Input

The first line of input contains two integer: n, m ($1 \leq n, m \leq 10^5$). The second line contains n integers, separated by space: $a[1], a[2], \dots, a[n]$ ($1 \leq a[i] \leq 10^9$) — initial value of array elements.

Each of the next m lines begins with a number $type$ ($type \in \{1, 2, 3\}$).

- If $type = 1$, there will be two integers more in the line: l, r ($1 \leq l \leq r \leq n$), which correspond the operation 1.
- If $type = 2$, there will be three integers more in the line:
 l, r, x ($1 \leq l \leq r \leq n; 1 \leq x \leq 10^9$), which correspond the operation 2.
- If $type = 3$, there will be two integers more in the line:
 k, x ($1 \leq k \leq n; 1 \leq x \leq 10^9$), which correspond the operation 3.

Output

For each operation 1, please print a line containing the answer. Notice that the answer may exceed the 32-bit integer.

Examples

Input	Output
5 5 1 2 3 4 5 2 3 5 4 3 3 5 1 2 5 2 1 3 3 1 1 3	8 5

Input	Output
10 10 6 9 6 7 6 1 10 10 9 5 1 3 9 2 7 10 9 2 5 10 8 1 4 7 3 3 7 2 7 9 9 1 2 4 1 6 6 1 5 9 3 1 10	49 15 23 1 9

Note

Consider the first testcase:

- At first, $a = \{1, 2, 3, 4, 5\}$.
- After operation 1, $a = \{1, 2, 3, 0, 1\}$.
- After operation 2, $a = \{1, 2, 5, 0, 1\}$.
- At operation 3, $2 + 5 + 0 + 1 = 8$.
- After operation 4, $a = \{1, 2, 2, 0, 1\}$.
- At operation 5, $1 + 2 + 2 = 5$.